

Example: The optimal tableau of the following problem is given below. Suppose that the objective function coefficient of x_1 is $9+\lambda$. Not to deteriorate the optimality condition, determine the range of λ .

$\max z = 9x_1 + 14x_2 + 5x_3$

$9x_1 + 4x_2 + 4x_3 \leq 54$

$9x_1 + 5x_2 + 5x_3 \leq 63$

$x_2 \leq 5$

$x_1, x_2, x_3 \geq 0$

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
1. z	0	0	0	0	1	9	108
λ x_1	1	0	0	5/9	-4/9	0	2
0 x_3	0	0	1	-1	1	-1	4
0 x_2	0	1	0	0	0	1	5

$$\max z = (9+\lambda)x_1 + 14x_2 + 5x_3$$

$\therefore$
 z -row $\Rightarrow$ Reduced cost for x_1, x_2, x_3 will be zero since they are basic in the opt. table.

Reduced cost for x_4 =

1	0	0
λ	5/9	$\frac{5}{9}\lambda$
0	-1	0
0	0	0

$$\boxed{\frac{5}{9}\lambda - 0}$$

R.C for x_5 =

1	1	1
λ	-4/9	$-\frac{4}{9}\lambda$
0	1	0
0	0	0

$$1 - \frac{4}{9}\lambda = 0 \Rightarrow \boxed{1 - \frac{4}{9}\lambda}$$

R.C for $x_6 =$

$$\begin{array}{r|rr} 1 & 9 & 9 \\ \lambda & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \end{array}$$

$$\boxed{9.} \geq 0!$$

$$\begin{array}{r|rr} 1 & 108 \\ \lambda & 2 \\ 0 & 4 \\ 0 & 5 \end{array}$$

$$\boxed{Z = 108 + 2\lambda}$$

$$\boxed{\frac{5}{9}\lambda \geq 0}$$

$$\lambda \geq 0$$

$$1 - \frac{4}{9}\lambda \geq 0$$

$$1 \geq \frac{4}{9}\lambda$$

$$\frac{9}{4} \geq \lambda$$

$$\boxed{0 \leq \lambda \leq \frac{9}{4}}$$